

ANNAI MIRA COLLEGE OF ENGINEERING AND TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY



SB8015 – NAAN MUDHALVAN CYBERSECURITY LABORATORY

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REGISTER NUMBER :

YEAR & BRANCH :

SEMESTER :

ACADEMIC YEAR :

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CERTIFICATE

This is to Certify that the Bonafide record of the practical work done by Register Number of III year **B.TECH INFORMATION TECHNOLOGY** submitted for the B.Tech - Degree practical examination (VI Semester) in **SB8015 – NAAN MUDHALVAN CYBERSECURITY LABORATORY** during the academic year **2025 –2026**.

STAFF IN-CHARGE

HEAD OF THE DEPARTMENT

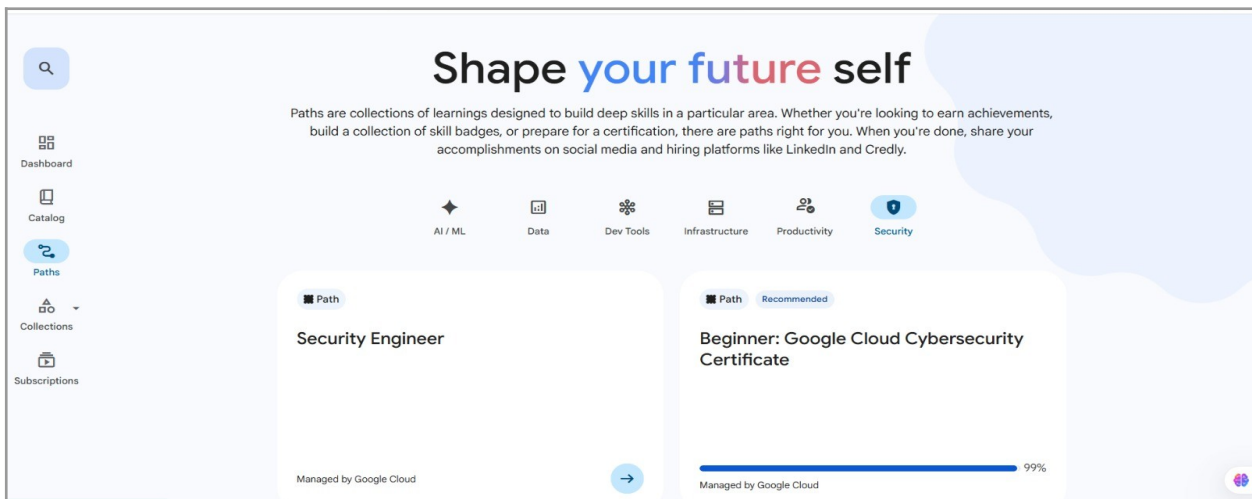
Submitted for the practical examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

Google Cloud Cybersecurity: Advanced Lab Manual

Welcome to the Under-Graduate (UG) level lab manual for Google Cloud Platform (GCP) Enterprise Cybersecurity. This advanced manual transitions from basic console operations to enterprise-grade security architecture, Infrastructure as Code (IaC), and automated threat remediation. This curriculum is designed for undergraduate students and security engineers, simulating a rigorous, hands-on enterprise environment.

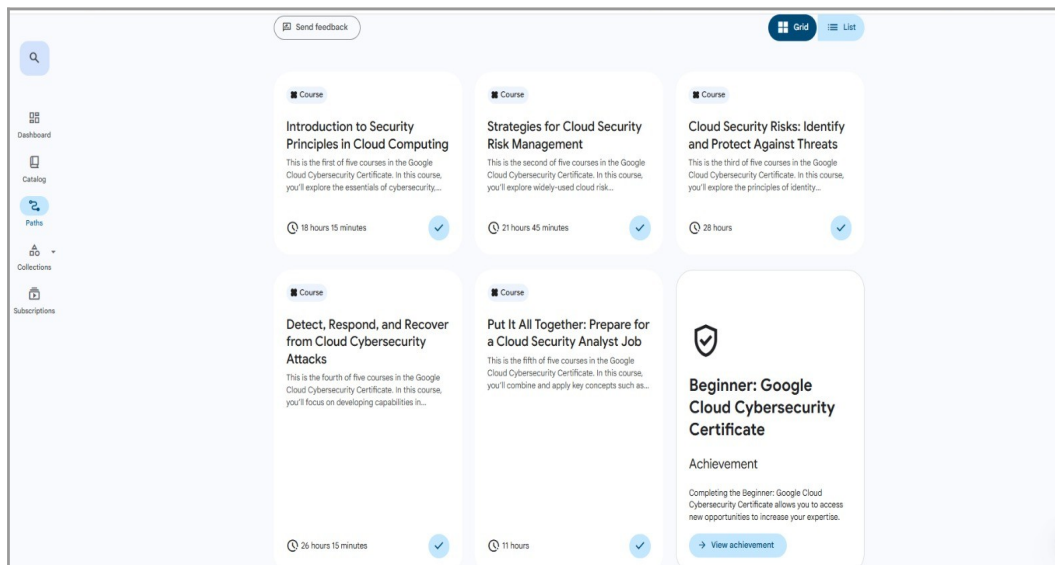


Overall Course Structure:

30 Advanced Lectures | 15 Research Papers/Case Studies | 5 Deep-Dive Assessments | 10 Complex Engineering Labs.

Table of Contents

- Module 1: Zero Trust & Advanced Identity Architecture
- Module 2: Enterprise Network Security & Perimeter Defense
- Module 3: Cryptography, HSMs, & Data Governance
- Module 4: Threat Hunting & Premium Security Operations
- Module 5: SOAR & Automated Incident Forensics



Module 1: Zero Trust & Advanced Identity Architecture

Module Breakdown:

6 Lectures | 3 Case Studies | 1 Assessment | 2 Labs.

This module moves beyond basic IAM to implement Zero Trust principles using contextual access, organizational policies, and keyless authentication models.

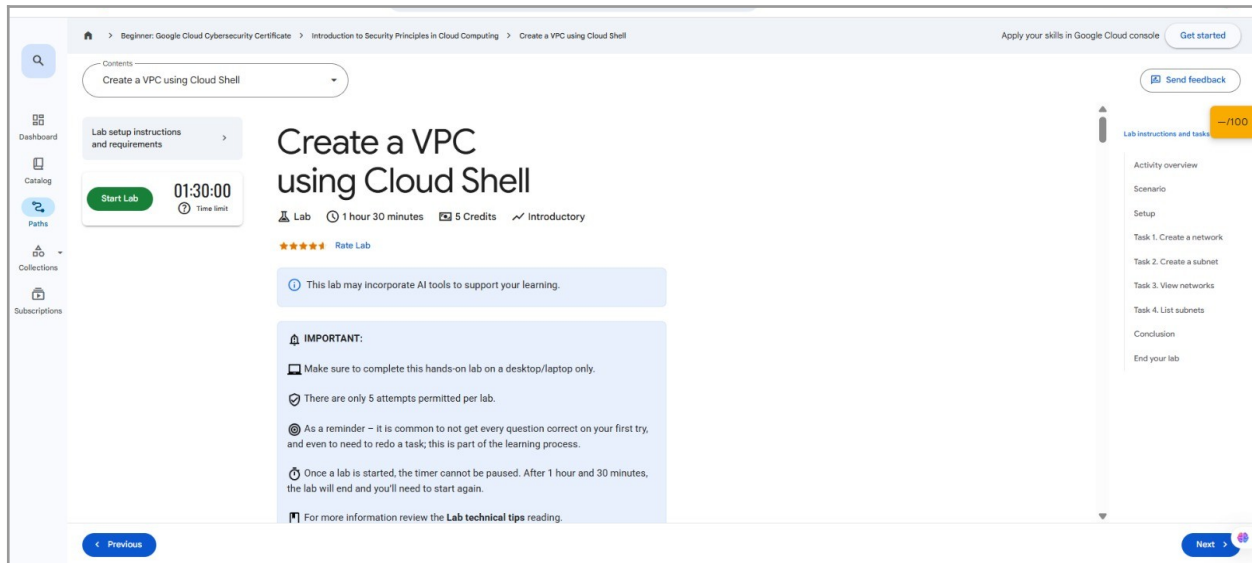
Lab 1.1: Implementing Workload Identity Federation

Element	Description
Difficulty	Advanced
Time	45 Minutes
Key Concepts	Workload Identity Pools, OIDC/SAML, Keyless Authentication

Objective:

Eliminate the use of long-lived Service Account JSON keys by configuring Workload

Identity Federation to allow an external AWS or GitHub Actions environment to authenticate to GCP dynamically.



Steps:

1. Open Cloud Shell and initialize the gcloud CLI.
2. Create a Workload Identity Pool: `gcloud iam workload-identity-pools create github-actions-pool --location="global"`.
3. Create an OIDC Provider within the pool tied to the external identity issuer (e.g., GitHub's OIDC URL).
4. Create a standard GCP Service Account (`deployer-sa`) with necessary deployment permissions.
5. Bind the external identity to the GCP Service Account by writing an IAM policy binding on the Service Account resource, mapping `principalSet://iam.googleapis.com/.../attribute.repository/` Person to the `roles/iam.workloadIdentityUser` role.
6. Test the keyless authentication pipeline from your external environment.

Lab 1.2: Enforcing Governance with Organization Policies

Element	Description
Difficulty	Advanced
Time	30 Minutes

Element	Description
Key Concepts	Org Node, Constraints, Resource Hierarchy

Objective:

Apply high-level security constraints across an entire GCP Organization to prevent configuration drift and shadow IT.

Steps:

1. Navigate to **IAM & Admin > Organization Policies**.
2. Locate the constraint: [constraints/compute.disableNestedVirtualization](#).
3. Edit the policy, override the parent configuration, and set the enforcement to **On** to mitigate hypervisor-level attacks.
4. Locate the constraint: [constraints/iam.disableServiceAccountKeyCreation](#).
5. Enforce this policy globally to ensure developers must use Workload Identity (configured in Lab 1.1) rather than exporting static credentials.

Lab 1.3: Contextual Access with BeyondCorp Enterprise

Element	Description
Difficulty	Advanced
Time	45 Minutes
Key Concepts	BeyondCorp Enterprise, Conditional Access, User Attributes, Custom Policies

Objective:

Implement a fine-grained, context-aware access policy for a critical internal application that requires users to be members of a specific Google Group AND connecting from an approved corporate network IP range.

Steps:

Deploy a sample internal web application in a GKE cluster or Compute Engine VM. Configure a basic Identity-Aware Proxy (IAP) setup for the application. Navigate to **Security > Access Context Manager**. Define a new **Access Level** that uses the logical operator **AND** to combine two conditions: **Source IP Subnet** (specifying a corporate VPN range) and **Required User Group** (specifying `security-admins@example.com`). Apply the newly created Access Level to the IAP resource securing the internal application. Test access: Verify that a user in the required group is denied access if they connect from an unauthorized public IP address.

Module 2: Enterprise Network Security & Perimeter Defense

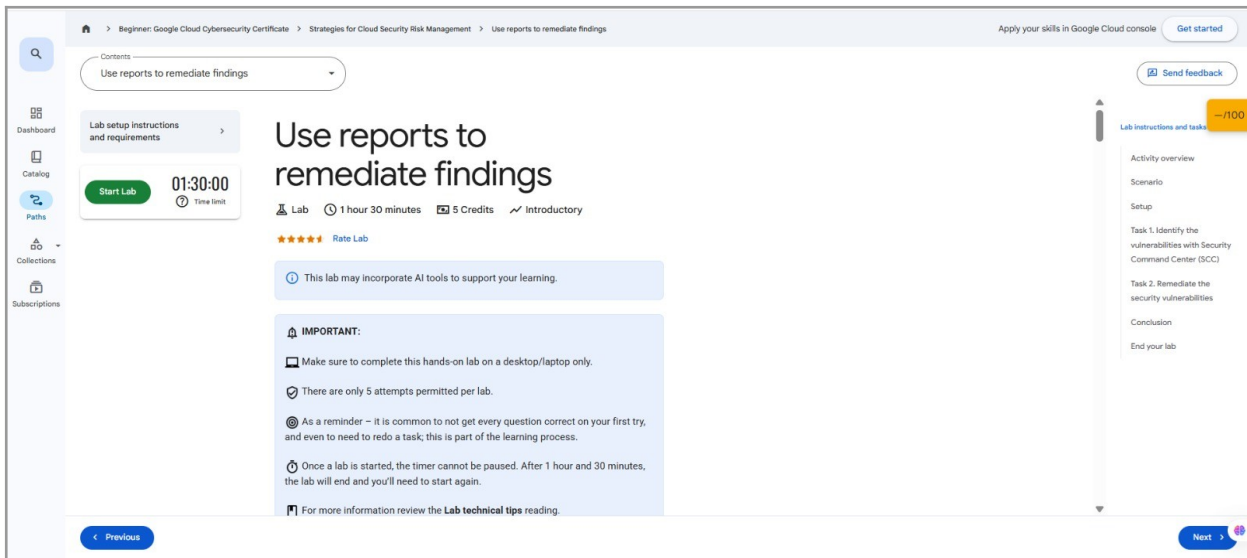
Module Breakdown:
6 Lectures | 3 Readings | 1 Assessment | 2 Labs.

Transition from single VPCs to enterprise topologies utilizing defense-in-depth, micro-segmentation, and API layer protection.

Lab 2.1: Constructing a Shared VPC with VPC Service Controls (VPC-SC)

Element	Description
Difficulty	Expert
Time	60 Minutes
Key Concepts	Host/Service Projects, Service Perimeters, Data Exfiltration Prevention

Objective:
Create a secure enclave that prevents data exfiltration even if IAM is misconfigured, by wrapping APIs in a VPC-SC perimeter.



Steps:

1. Set up a **Host Project** and designate two **Service Projects** (e.g., **data-lake-prod** and **app-engine-prod**).
2. Configure a Shared VPC in the Host project and attach the Service projects.
3. Navigate to **Security > VPC Service Controls**.
4. Create an Access Policy scoped to your organization.
5. Create a **Service Perimeter**. Add the **data-lake-prod** project to the perimeter.
6. Restrict the **storage.googleapis.com** and **bigquery.googleapis.com** services.
7. Attempt to access a BigQuery dataset inside the perimeter from an external, unauthorized IP. Verify that the request is denied by VPC-SC despite having IAM permissions.

Lab 2.2: Context-Aware Access and Cloud Armor WAF

Element	Description
Difficulty	Advanced
Time	45 Minutes
Key Concepts	BeyondCorp, WAF, Layer 7 Defenses, Access Levels

Objective:

Protect a public-facing application using Google Cloud Armor for L7 filtering and Identity-Aware Proxy (IAP) with device-based context.

Steps:

1. Deploy a sample web application behind an External HTTPS Load Balancer.
2. Navigate to **Security > Access Context Manager**.
3. Create an **Access Level** that requires incoming connections to be from a specific IP subnet AND from a device with an encrypted hard drive (requires Endpoint Verification extension).
4. Enable **IAP** on the backend service and apply the Access Level.
5. Navigate to **Network Security > Cloud Armor**.
6. Create a security policy. Add a rule to evaluate incoming traffic against pre-configured WAF rules (cve-canary, sqli-v33, xss-v33).
7. Attach the Cloud Armor policy to the Load Balancer's backend service. Simulate an SQL injection attack and verify it is blocked with a 403 Forbidden.

Lab 2.3: Hardened GKE with Private Service Connect (PSC)

Element	Description
Difficulty	Expert
Time	75 Minutes
Key Concepts	GKE Private Cluster, Private Service Connect, Control Plane Visibility

Objective:

Deploy a private Google Kubernetes Engine (GKE) cluster where the control plane is secured via Private Service Connect (PSC) instead of a publicly routable endpoint, minimizing network attack surface.

Steps:

Create a GKE Private Cluster in a new VPC network. Ensure **Private cluster** is

enabled and **Enable Private Service Connect** is selected for the control plane endpoint.

Verify the cluster creation and note that the control plane's endpoint is a private IP address and the Master global access is disabled.

Attempt to connect to the cluster's control plane (using **kubectl**) from a VM in a different, non-peered VPC network. Verify the connection is denied.

Configure a **Private Service Connect Endpoint** in the second VPC that targets the GKE Control Plane Service Attachment.

Update the **kubectl** configuration on the second VM to use the PSC endpoint's IP address.

Verify that **kubectl** access is now successfully established from the second, isolated VPC network via the private PSC connection.

Module 3: Cryptography, HSMs, & Data Governance

Module Breakdown:

6 Lectures | 3 Research Papers | 1 Assessment | 2 Labs.

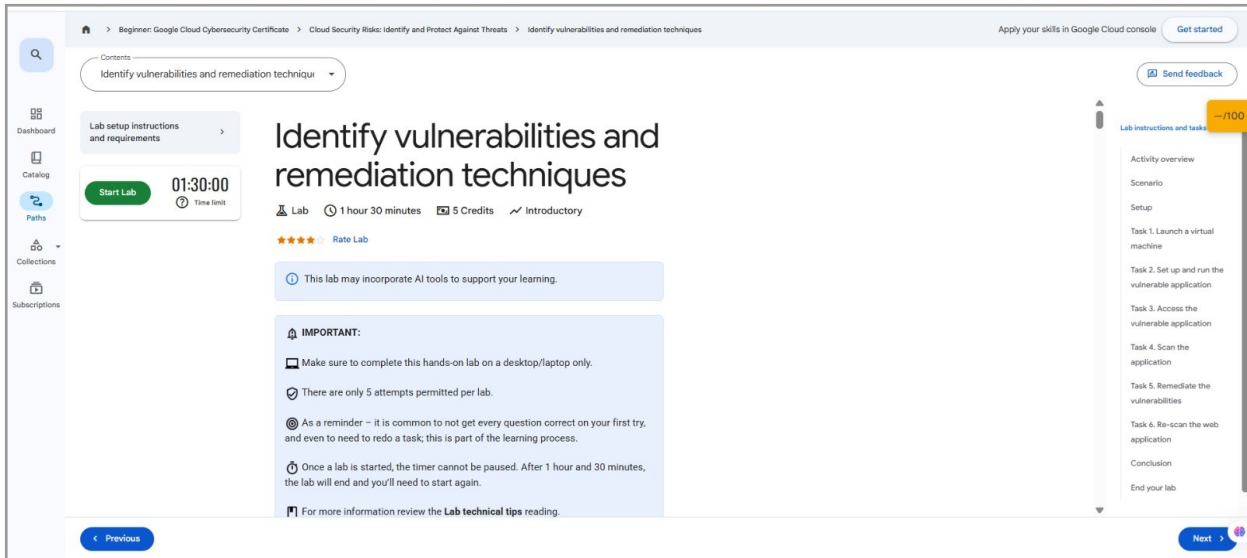
Focus on regulatory compliance, hardware-backed cryptography, and automated sensitive data discovery.

Lab 3.1: Automated PII Discovery with Cloud DLP

Element	Description
Difficulty	Advanced
Time	45 Minutes
Key Concepts	Data Loss Prevention, Inspection Templates, De-identification

Objective:

Automatically scan unstructured data in Cloud Storage for Personally Identifiable Information (PII) and mask it.



Steps:

1. Upload a sample dataset containing raw credit card numbers and Social Security Numbers to a Cloud Storage bucket.
2. Navigate to **Security > Data Loss Prevention**.
3. Create an **Inspection Template**. Select **CREDIT_CARD_NUMBER** and **US_SOCIAL_SECURITY_NUMBER** as the infoTypes.
4. Create a **De-identification Template** to mask the discovered infoTypes using a cryptographic token (Format Preserving Encryption).
5. Set up a DLP **Job Trigger** that scans new objects uploaded to the bucket, utilizing the inspection and de-identification templates to output a sanitized version of the file to a secure destination bucket.

Lab 3.2: Provisioning Hardware-Backed Keys with Cloud HSM

Element	Description
Difficulty	Advanced
Time	30 Minutes
Key Concepts	FIPS 140-2 Level 3, Hardware Security

Element	Description
	Modules, Asymmetric Signing

Objective:

Generate and use cryptographic keys strictly bound to a FIPS 140-2 Level 3 certified hardware security module for highly regulated workloads.

Steps:

1. Open Cloud Shell.
2. Create a KMS Key Ring: `gcloud kms keyrings create hsm-ring --location=us-central1`.
3. Create an HSM-backed key: `gcloud kms keys create root-signing-key --keyring=hsm-ring --location=us-central1 --purpose=asymmetric-signing --default-algorithm=ec-sign-p256-sha256 --protection-level=hsm`.
4. Retrieve the public key to distribute to verifiers.
5. Create a sample digital contract (text file) and use the `gcloud kms asymmetric-sign` command to generate a cryptographic signature using the HSM hardware.
6. Download the attestation statement provided by Google verifying the key material never leaves the HSM hardware.

Lab 3.3: Deploying a Confidential VM for Attested Workloads

Element	Description
Difficulty	Advanced
Time	30 Minutes
Key Concepts	Confidential Computing, vTPM, Remote Attestation, Measured Boot

Objective:

Deploy a Confidential Compute Engine VM and verify its security posture using remote attestation to ensure the virtual hardware and kernel have not been tampered with before processing sensitive data.

Steps:

Create a new Compute Engine VM instance. Select an appropriate machine type (e.g., N2D or C2D) and enable the **Confidential VM service** option.

Enable **Secure Boot** and **vTPM** for the instance, ensuring the VM's boot process is measured.

Once the VM is running, retrieve the initial boot-time attestation report using the `gcloud compute instances get-shielded-identity` command.

Simulate a state change (e.g., manually modify a kernel parameter) on the VM.

Generate a new attestation report and compare the PCR values with the initial report.

Analyze the report differences to determine if the integrity measurements have been violated, confirming the utility of remote attestation for detecting tampering.

Module 4: Threat Hunting & Premium Security Operations

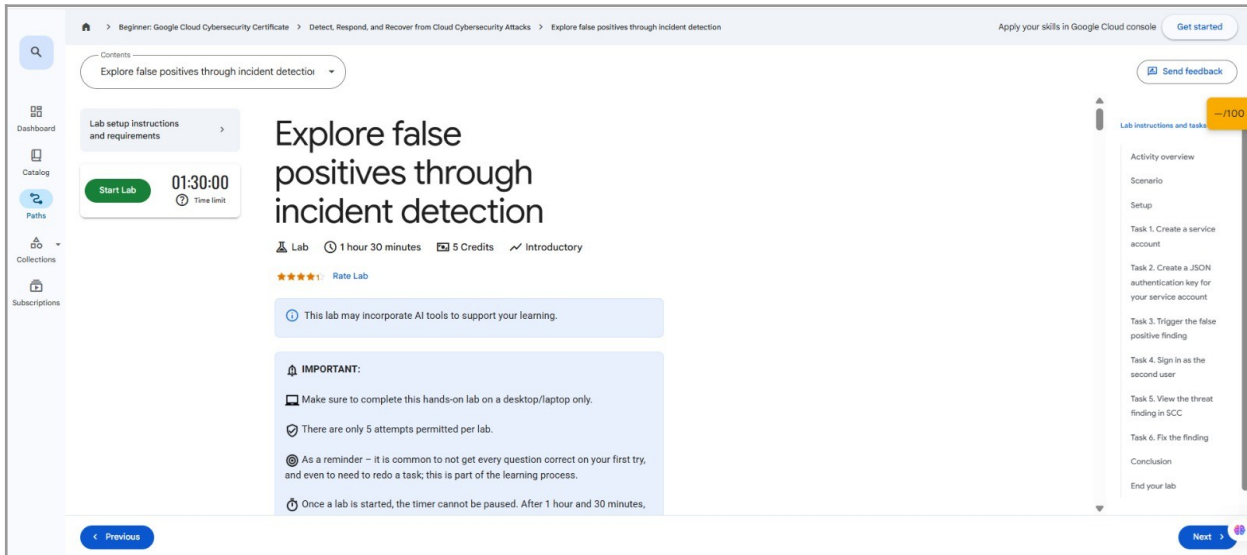
Module Breakdown:
6 Lectures | 3 Readings | 1 Assessment | 2 Labs.

Utilize advanced log analytics, BigQuery for SIEM-like capabilities, and Security Command Center Premium features.

Lab 4.1: Threat Hunting via VPC Flow Logs and BigQuery

Element	Description
Difficulty	Advanced
Time	50 Minutes
Key Concepts	Flow Logs, Log Sinks, SQL Data Analysis, IOC matching

Objective:
Export network telemetry to a data warehouse to execute complex SQL queries hunting for Indicators of Compromise (IOCs) and beaconing behavior.



Steps:

1. Enable **VPC Flow Logs** on your primary subnet, setting the aggregation interval to 5 seconds and including metadata.
2. In **Log Router**, create a sink to route all compute.googleapis.com/vpc_flows to a new **BigQuery dataset** ([secops_telemetry](#)).
3. Generate simulated malicious traffic (e.g., executing Nmap scans or establishing a reverse shell) between two VMs.
4. Open the **BigQuery Workspace**.
5. Write an advanced SQL query to group network connections by source IP and destination port, identifying anomalies such as a single internal IP attempting connections across 1,000+ distinct ports within a 1-minute window (detecting lateral movement/port scanning).

Lab 4.2: SCC Premium: Event Threat Detection & Custom Modules

Element	Description
Difficulty	Expert
Time	40 Minutes

Element	Description
Key Concepts	ETD, Custom Threat Modules, YARA rules

Objective:

Leverage Security Command Center Premium to detect advanced threats in real-time based on log streams and custom logic.

Steps:

1. Navigate to **Security Command Center > Settings > Event Threat Detection**.
2. Review active ETD rules (e.g., Cryptomining, Malware Domain Execution, Brute Force SSH).
3. Navigate to **Security Health Analytics** and select **Custom Modules**.
4. Create a custom module using the CEL (Common Expression Language) syntax to flag any Cloud Storage bucket created without CMEK encryption enabled.
5. Intentionally violate the policy by creating a non-compliant bucket, and observe the High-Severity finding populate in the SCC dashboard in real-time.

Lab 4.3: Automating Findings Triage with Cloud Workflows

Element	Description
Difficulty	Expert
Time	60 Minutes
Key Concepts	Security Command Center API, Cloud Workflows, Pub/Sub, Ticketing System Integration

Objective:

Build a serverless orchestration workflow that automatically subscribes to high-severity Security Command Center (SCC) findings, enriches the finding data, and automatically creates a Jira or ServiceNow ticket for immediate triage.

Steps:

Configure a **Notification Export** in SCC to send all high-severity **IAM_ANOMALY** findings to a dedicated Pub/Sub topic (e.g., **scc-high-severity**).

Develop a Cloud Workflow definition (using YAML or JSON) that is triggered by the Pub/Sub message.

The workflow must perform three sequential steps:

Step 1: Parse the SCC finding JSON payload.

Step 2: Call an internal Cloud Function endpoint to enrich the finding (e.g., look up the Project Owner's email).

Step 3: Call an external ticketing system API (simulated or real) to create a new ticket, populating the summary and description with the enriched data.

Simulate an IAM anomaly finding (e.g., granting a user a Project Owner role) and verify that the Cloud Workflow executes successfully, leading to the creation of an automated ticket.

Module 5: SOAR & Automated Incident Forensics

Module Breakdown:

6 Lectures | 3 Case Studies | 1 Assessment | 2 Labs.

Shift from reactive monitoring to Security Orchestration, Automation, and Response (SOAR), writing code to automatically contain threats.

Lab 5.1: Building a Serverless Automated Remediation Pipeline

Element	Description
Difficulty	Expert
Time	60 Minutes
Key Concepts	Eventarc, Cloud Functions, IAM Automation, SOAR

Objective:

Write a Python function that automatically strips permissions from a user if they are detected performing dangerous actions.

Steps:

1. In **Pub/Sub**, create a topic named **iam-violations**.
2. In **Log Router**, create a sink that filters for **protoPayload.methodName="v1.compute.firewalls.delete"** (detecting someone deleting network defenses) and sends it to the **iam-violations** topic.
3. Navigate to **Cloud Functions** and create a 2nd Gen Python 3 function triggered by Eventarc listening to your Pub/Sub topic.
4. **Coding Task:** Write a Python script using the **google-cloud-iam** and **google-cloud-resource-manager** client libraries. The function must extract the **principalEmail** from the incoming log payload, and automatically remove their IAM bindings from the project using the REST API.
5. Deploy the function. Test it by having a test account Person delete a firewall rule, and watch the system automatically revoke their access within seconds.

Lab 5.2: Automated Forensic Disk Snapshotting

Element	Description
Difficulty	Advanced
Time	45 Minutes
Key Concepts	Digital Forensics, Disk Snapshots, Network Isolation

Objective:

Create an automated runbook that isolates a compromised Virtual Machine and captures its memory/disk state for forensic analysis.

Steps:

1. Deploy a "target" VM simulating a production workload.
2. Simulate a high-severity SCC alert regarding this VM (e.g., Malware Execution detected).
3. Using Cloud Shell or a Cloud Function pipeline, execute the forensic containment script:
 - **Step A (Isolation):** Use **gcloud compute instances add-tags** to apply an

isolated tag to the VM.

- **Step B (Containment):** Ensure a pre-existing firewall rule exists that explicitly denies *all* egress/ingress traffic for the isolated tag.
 - **Step C (Preservation):** Use `gcloud compute disks snapshot` to take an immediate, read-only snapshot of the VM's boot disk.
4. Mount the newly created forensic snapshot as a secondary disk to a separate, isolated "Forensics Investigator" VM to analyze the malware without risking further contamination.

Lab 5.3: Infrastructure as Code (IaC) Policy Enforcement

Element	Description
Difficulty	Advanced
Time	45 Minutes
Key Concepts	Policy Controller, Gatekeeper, Kubernetes Security Context, Admission Control

Objective:

Utilize Policy Controller (powered by Open Policy Agent Gatekeeper) to enforce security best practices at the Kubernetes cluster admission layer, preventing developers from deploying non-compliant infrastructure.

Steps:

Enable **Policy Controller** on an existing GKE cluster.

Install a **Constraint Template** for a specific security rule (e.g., requiring all Pods to set a `securityContext` and forbidding `allowPrivilegeEscalation`).

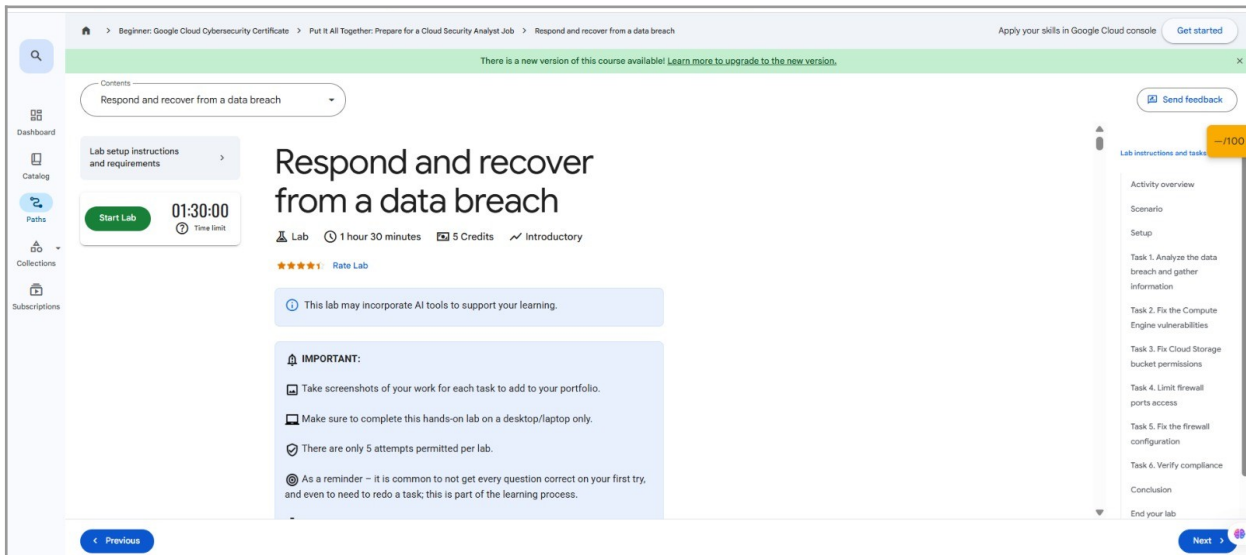
Deploy a corresponding **Constraint** resource that utilizes the template to enforce the policy cluster-wide.

Attempt to deploy a non-compliant Kubernetes manifest (e.g., a Pod specification without a `securityContext` defined).

Verify that the Admission Controller blocks the deployment with a 403 Forbidden error and a detailed error message explaining the policy violation, demonstrating preventive enforcement.

Google Cloud Certification Pathway: Professional Cloud Security Engineer

This program serves as the industry standard for validating the advanced knowledge gained in this manual. It is designed to certify security engineers who can design, develop, and manage a secure infrastructure on Google Cloud Platform (GCP).



Certification Domains & Alignment:

The exam domains directly align with the architecture and topics covered in this lab manual:

- **Designing and implementing a secure network:** Aligns with Module 2 (Enterprise Network Security & Perimeter Defense).
- **Managing identity and access:** Aligns with Module 1 (Zero Trust & Advanced Identity Architecture).
- **Ensuring data protection:** Aligns with Module 3 (Cryptography, HSMs, & Data Governance).
- **Managing operations and risk:** Aligns with Module 4 (Threat Hunting) and Module 5 (SOAR & Automated Incident Forensics).

Recommended Foundational Skill Badges:

To ensure a strong foundation before undertaking the labs in this advanced manual, completing the following Google Cloud Skill Badges is highly recommended:

- Security in Google Cloud
- Secure Cloud Infrastructure

Next Steps:

Obtaining this certification validates the real-world skills developed through these labs, positioning the user as a Professional Cloud Security Engineer.

Expanded Lab Command Procedures and Visual Aids

This section provides detailed command breakdowns and visualizations for key labs requiring Google Cloud CLI (gcloud) interaction.

Module 1: Zero Trust & Advanced Identity Architecture

Lab 1.1: Implementing Workload Identity Federation - Command Breakdown

Step	Procedure/Command	Description
1-2	<code>gcloud iam workload-identity-pools create github-actions-pool --location="global"</code>	Initializes a global identity pool named `github-actions-pool` to manage external identities, eliminating the need for static keys.
3-5	(Requires Console/API)	Create the OIDC provider (linked to external issuer), then the GCP Service Account (`deployer-sa`), and finally apply the IAM policy binding to map the external identity to the Service Account.
6	(External Environment)	Use the configured identity in your external environment (e.g., GitHub Actions) to obtain a short-lived token and authenticate to GCP dynamically.

Visual Aid: Workload Identity Federation Keyless Flow

Module 3: Cryptography, HSMs, & Data Governance

Lab 3.2: Provisioning Hardware-Backed Keys with Cloud HSM - Command Breakdown

Step	Procedure/Command	Description
2	<code>gcloud kms keyrings create hsm-ring --location=us-central1</code>	Creates a Key Ring to logically group the keys within a specific region.
3	<code>gcloud kms keys create root-signing-key --keyring=hsm-ring --location=us-central1 --purpose=asymmetric-signing --default-algorithm=ec-sign-p256-sha256 --protection-level=hsm</code>	Creates an asymmetric key specifically bound to the FIPS 140-2 Level 3 certified Cloud HSM.
5	<code>gcloud kms asymmetric-sign --key=root-signing-key --keyring=hsm-ring --location=us-central1 --digest-path=contract_hash.sha256 --input-file=contract.txt --output-file=signature.bin</code>	Uses the HSM key to digitally sign a document hash, proving the key material never left the hardware.

Lab 3.3: Deploying a Confidential VM for Attested Workloads - Command Breakdown

Step	Procedure/Command	Description
1-2	(Requires Console/gcloud)	Deploy the VM with the `--confidential-compute` flag enabled and `--shielded-secure-boot` and `--shielded-vtpm` options set to enable measured boot.
3	<code>gcloud compute instances get-shielded-identity INSTANCE_NAME --zone=ZONE --format=json</code>	Retrieves the initial, trusted state of the VM's virtual Trusted Platform Module (vTPM) Platform Configuration Registers

		(PCRs).
5	<pre>gcloud compute instances get- shielded-identity INSTANCE_NAME --zone=ZONE --format=json</pre>	Re-runs the command after a simulated tamper to compare the new PCR values against the initial report, validating integrity.

Module 5: SOAR & Automated Incident Forensics

Lab 5.2: Automated Forensic Disk Snapshotting - Command Breakdown

Step	Procedure/Command	Description
3A	<pre>gcloud compute instances add- tags TARGET_VM_NAME -- tags=isolated-for-forensics -- zone=ZONE</pre>	Adds a specific network tag to the target VM to immediately trigger an existing firewall rule for network isolation (Containment).
3C	<pre>gcloud compute disks snapshot TARGET_DISK_NAME --snapshot- names=forensic-capture-\$(date + %Y%m%d-%H%M%S) -- zone=ZONE --storage- location=GLOBAL</pre>	Creates a non-modifiable, read-only snapshot of the disk at the moment of compromise for preservation and analysis.

Google Cloud Cybersecurity certificate:

